SRI CHAITANYA EDUCATIONAL INSTITUTIONS

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

NEET GRAND TEST-1

Name :..... Hall Ticket No: | | | | | | | |

INSTRUCTIONS TO CANDIDATES

1. The Model **NEET- 2020** is of **3 Hrs** duration. **Time: 10.00 AM – 1.00 PM**.
2. The question paper for NEET-2020 consists of 180 questions comprising 45 questions in Botany, 45 in Zoology, 45 in Physics and 45 in Chemistry for NEET.
3. All questions are of objective type (Multiple choices only)
4. Each question carries four marks.
5. Negative marking: one mark will be deducted for every wrongly answered question.
6. Total Marks 720.
7. The candidates are prohibited from carrying any paper to the examination hall except **HALL TICKET**.
8. No Calculators, Mini-Cards, Watches with Calculators, Pager, Cell Phone, Slide rules or outer aids to calculation will be allowed in the examination hall.
9. Candidates once admitted will not be allowed to leave the hall till half an hour before the closing of the test.
10. A separate sheet is attached in the middle of this booklet for rough work, you can detach and use it.
11. A detachable answer sheet with 180 question blocks, with 4 circles corresponding to 4 multiple choice for each question will be provided. Use HB Pencil to darken the appropriate circle against the question number provided in the sheet. Answer should be marked only on the answer sheet, but not on the question paper booklet.

1. Taylor and colleagues proved that the DNA in chromosomes replicates semiconservatively by conducting experiments on
   (1) *E.coli*
   (2) *Drosophila*
   (3) *Vicia faba*
   (4) Bacteriophage
2. Which one of the following is synthesised constitutively in the Lac operon ?
   (1) Inducer
   (2) Repressor
   (3) β-galactosides
   (4) Transacetylase
3. The Roquefort cheese is ripened by using a specific
   (1) Fungus
   (2) Bacterium
   (3) Yeast
   (4) Actinomycetes
4. How many phenotypes are present for the human ABO blood types ?
   (1) 9
   (2) 3
   (3) 4
   (4) 6
5. Smooth endoplasmic reticulum is
   (1) Site of active protein synthesis
   (2) Site of formation of glyco proteins and glycolipids
   (3) Major site for the synthesis of lipids
   (4) Involved in secretion
6. Osculum is an opening of
   (1) Hydra, acts as both mouth and anus
   (2) Tapeworm, to remove the nitrogenous waste
   (3) Sponge, helps in outflow of water
   (4) Ctenophore, to expel the gametes
7. In which of the following animals, metamorphosis takes place in freshwater?
   (1) *Obelia*
   (2) *Musca*
   (3) *Bombyx*
   (4) *Petromyzon*
8. Which set is not correctly matched?

Name of the animal	Salient features
(1) *Calotes*	a) Fertilization is internal b) Development is external
(2) *Periplaneta*	a) Bilateral symmetry b) Chitinous exoskeleton
(3) *Struthio*	a) Pneumatic bones b) Well - developed flight muscles
(4) *Asterias*	a) Radial symmetry b) Radial cleavage

9. Body surface helps in absorption of digested food in
   (1) Jellyfish
   (2) Tapeworm
   (3) Roundworm
   (4) Silverfish

10. Epithelium in tubular part of nephron is mostly

(1) Squamous epithelium

(2) Columnar epithelium

(3) Cuboidal epithelium

(4) Pseudostratified epithelium

11. Specialised cells in fixing nitrogen in cyanobacteria are

(1) Heterocysts

(2) Hormogonia

(3) Akinetes

(4) Idioblast

12. Which of these statements is incorrect ?

(1) Mg^{2+} is an activator of pyruvate dehydrogenase

(2) Citrate is isomerised as isocitrate in TCA cycle

(3) The continued oxidation of acetyl CoA via TCA cycle requires replenishment of oxaloaceticacid and regeneration of NAD^+ and FAD from $NADH + H^+$ and $FADH_2$

(4) Cytochrome C is a mobile electron carrier between complex II and complex III

13. Lamp brush chromosomes occur in

(1) Mitotic prophase stage of salivary glands of insects

(2) Diplotene stage of most animal oocytes

(3) Suspensor cells of embryosac

(4) Pachytene stage of animal oocytes and spermatocytes

14. In the cell walls of diatoms _____ is present

(1) Cutin

(2) Suberin

(3) Silica

(4) Murein

15. Which structure is absolutely necessary for many biological activities of proteins ?

(1) Quaternary structure

(2) Primary structure

(3) Secondary structure

(4) Tertiary structure

16. Cells of the following connective tissue do not form fibers

(1) Bone

(2) Cartilage

(3) Blood

(4) All of these

17. Arrangement of setae in earthworm as "ring of setae" is termed

(1) Peripatetics

(2) Peritrichous

(3) Peripherals

(4) Perichaetine

18. In cockroach a series of segmentally arranged ganglia are joined by paired

(1) Longitudinal connectives

(2) Horizontal connectives

(3) Longitudinal commissures

(4) Horizontal commissures

19. Teeth in man with 3 roost each are

(1) Incisors

(2) Canines

(3) Premolars

(4) Molars

20. Which of the following statement is not true?

(1) Lysozyme is an antibacterial agent

(2) Interferon is an antiviral protein

(3) Heparin is an anticoagulant

(4) Histamine is an antifungal agent

21. Which of the following is not a function of accessory pigments ?

(1) Absorption of light and transfer energy to chlorophyll a

(2) Enable a wide range of wavelengths of incoming light to be utilised for photosynthesis

(3) Protect chlorophyll a from photo-oxidation

(4) Proton translocation across the membrane

22. Which is not an external factor that affect rate of transpiration ?

(1) Temperature

(2) Wind speed

(3) Humidity

(4) Canopy structure

23. Which region of Golgi apparatus is involved in the modification of proteins synthesised by ribosomes on the endoplasmic reticulum ?

(1) Cis face

(2) Trans face

(3) Cisternae

(4) Cristae

24. Correct match regarding amount of DNA during different stages of meiosis

Meiocyte at GI phase	Daughter nuclei at the end of meiosis I	Daughter nuclei at end of meiosis II
(1) 2C	2C	1C
(2) 4 C	2C	1C
(3) 4C	2C	2C
(4) 2C	4C	2C

25. Starch sheath is

(1) Endodermis of root

(2) Endodermis of monocot stem

(3) Bundle sheath of dicot leaf

(4) Endodermis of dicot stem

26. Identify the correct combination by taking their maximum volumes

List-I	List-II	List-III
(1) IC	RV+IRV	3500 mL
(2) EC	TV+ERV	1300 mL
(3) FRC	ERV+RV	2300 mL
(4) VC	ERV + IRV + RV	4600 mL

27. When O^{+ve} man marries AB^{-ve} woman, what is the probability that an O^{+ve} child with HDNB is born in the second conception of that couple? (Father is heterozygous for Rh factor)

(1) 25%

(2) 50%

(3) 0%

(4) 75%

28. Read the following
(a) Decrease the rate of heartbeat
(b) Decrease the speed of conduction of action potentials
(c) Decrease in cardiac output
The above functions/effects related to
(1) Epinephrine secreted by adrenal medulla
(2) Sympathetic division
(3) Parasympathetic division
(4) Steroid hormones secreted by adrenal cortex

29. Which of the following is a feature of descending limb of loop of Henle?
(1) Highly permeable to electrolytes
(2) Permeable to water
(3) Impermeable to water
(4) Totally impermeable electrolytes

30. Match the following

Column-I	Column-II
(a) Protonephridia	(i) Leech
(b) Nephridia	(ii) Rotifer
(c) Antennal glands	(iii) Prawn
(d) Proboscis glands	(iv) Acorn worm

(1) (a)-(i); (b)-(ii); (c)-(iii); (d)-(iv)
(2) (a)-(ii); (b)-(i); (c)-(iii); (d)-(iv)
(3) (a)-(i); (b)-(ii); (c)-(iv); (d)-(iii)
(4) (a)-(ii); (b)-(i); (c)-(iv); (d)-(iii)

31. A typical mature angiosperm embryosac is
(1) 7-nucleated, 8-celled
(2) 8-nucleated, 7-celled
(3) 8- nucleated, 8-celled
(4) 7-nucleated, 7-celled

32. The trans membrane channel F_o transports protons across the membrane through
(1) Simple diffusion
(2) Active transport
(3) Facilitated diffusion
(4) Either by active or passive transport

33. Pollen grains are well preserved as fossils because of the presence of
(1) Sporopollenin
(2) Pecto cellulose
(3) Ligno suberin
(4) Chitin

34. An element required for uptake and utilisation of Ca^{2+} is
(1) Iron
(2) Manganese
(3) Potassium
(4) Boron

35. *Nitrobacter* is a
(1) Chemoheterotroph
(2) Chemo autotroph
(3) Photo heterotroph
(4) Photo autotroph

36. Identify the mismatched combination from the following
(1) Troponin – Thin filament
(2) Meromyosin- Thick filament
(3) Zygomatic bone – Facial bone
(4) Glenoid cavity – Pelvic girdle

37. The meninx that is in contact with the brain tissue is
(1) Piamater
(2) Grey matter
(3) Duramater
(4) White matter

38. Gull's disease is a/an

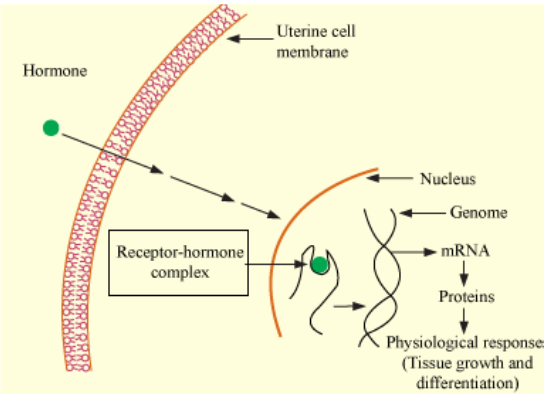
(1) Autoimmune disease

(2) Hormone deficiency disease

(3) Nutritional deficiency disease

(4) Contagious disease

39. Study the following diagram



In the above hormonal mechanism, if 'X' is receptor and 'Y' is hormone in receptor-hormone complex, then 'X' and 'Y' are usually

(1) X-Protein; Y-Lipid

(2) X-Protein; Y-Glycoprotein

(3) X-Lipid; Y-Protein

(4) X-Lipid; Y-Lipid

40. Macula of ear and macula of eye respectively are

(1) Part of otolith organ; Yellow spot

(2) Base of semicircular canals; Yellow spot

(3) Part of otolith organ; Part with only cones

(4) Basal part of semicircular canals; Part with only cones

41. Correct match the following

	List-I	List-II
A.	Polyethylene glycol	I. Encapsulation of somatic embryoids
B.	Ethidium bromide	II. Fusogen
C.	*Taq* polymerase	III. Visualising DNA fragments
D.	Sodium alginate	IV. Thermostable DNA polymerase

	A	B	C	D
(1)	I	III	IV	II
(2)	II	III	IV	I
(3)	II	I	IV	III
(4)	II	IV	I	III

42. Incorrect match of the following

(1) Alternate phyllotaxy, false septum, dry fruit – mustard

(2) Trimerous flowers, axile placentation, fleshy fruit - *Asparagus*

(3) Cymose inflorescence, imbricate aestivation, fleshy fruit – sun hemp

(4) Cymose inflorescence, adhesion of stamens, oblique septa - *Solanum*

43. Phenotype ratio is 1 : 2 : 1 in

(1) Shape of fruit in pea

(2) Flower colour in snapdragon

(3) Shape of seeds in *Pisum*

(4) Colour of pod in pea

44. Incorrect match of the following

(1) Heterogametes - *Fucus*

(2) Dioecious plant – Date palm

(3) Homogametes - *Cladophora*

(4) Monoecious gametophyte – *Marchantia*

45. Which phytohormone promotes growth of new leaves, chloroplasts in leaves, lateral shoot growth and adventitious shoot formation ?
(1) Auxins
(2) Gibberellins
(3) Ethylene
(4) Cytokinins

46. Cocaine is a
(1) Depressant
(2) Stimulant
(3) Resin
(4) Tannin

47. Frog differs from man by
(1) Having menstrual cycle
(2) Having metamorphosis
(3) Having oestrous cycle
(4) Having morphogenesis

48. Primary oocyte is surrounded by a single layer of granulosa cells in
(1) Primary follicle
(2) Secondary follicle
(3) Tertiary follicle
(4) Graafian follicle

49. Types of cells that line the seminiferous tubules are
(1) Male germ cells and Leydig cells
(2) Sertoli cells and Leydig cells
(3) Male germ cells and Sertoli cells
(4) Male germ cells, Sertoli cells and Leydig cells

50. First movement of foetus takes place during
(1) 4th month
(2) 5th month
(3) 6th month
(4) 7th month

51. Round up is
(1) Transgenic tomato
(2) Pest resistant flat bean
(3) Broad spectrum herbicide
(4) Male sterile mustard

52. Which one of the following is an example for molecular farming ?
(1) Pest resistant Bt cotton
(2) Vitamin A enriched rice
(3) Atlas 66 having high protein content
(4) Hirudin production by transgenic *Brassica napus*

53. Both autogamy and geitonogamy is prevented in
(1) *Commelina*
(2) Castor
(3) Maize
(4) Papaya

54. Use of bio resources by multinational companies and organisations without authorisation from the concerned country and it's people is called
(1) Bio exploitation
(2) Biodegradation
(3) Bio-infringement
(4) Biopiracy

55. Zygotic meiosis is characteristic of
(1) *Marchantia*
(2) *Fucus*
(3) *Funaria*
(4) *Chlamydomonas*

56. In humans the development starts in
(1) Ovary
(2) Uterus
(3) Fallopian tube
(4) Graafian follicle

57. Lichen, Fig-Wasp, Mycorrhiza, *Ophrys*-Bumblebee, Epiphyte–Mango, Barnacle – Whale, Sea anemone – Clownfish
How many of above showing mutualism?
(1) Four
(2) Five
(3) Six
(4) Two

58. Identify the incorrect combination from the following
(1) Periodic abstinence – Natural method
(2) Cervical caps – Barrier method
(3) Lippes loop – Hormonal IUD
(4) Saheli – OCPs

59. In test tube baby programme the development of embryo after 16 blastomere stage is done in
(1) Hysterectomized biological mother
(2) Glass jar
(3) Test tube
(4) Surrogate mother

60. Spermatogenesis takes place by mitotic divisions only in
(1) *Aedes*
(2) *Apis*
(3) *Aphid*
(4) *Apus*

61. Charging of t – RNA refer to
(1) Binding of ATP to tRNA
(2) Linking of activated amino acids to tRNA
(3) Binding of tRNA to mRNA
(4) Formation of peptide bond between tRNA

62. In mungbean resistance to yellow mosaic virus is introduced by
(1) Genetic engineering
(2) Mutation breeding
(3) Hybridisation, selection
(4) Selection amongst somaclonal varients

63. Convolulaceae and solanaceae belong to order namely
(1) Polymoniales
(2) Sapindales
(3) Poales
(4) Parietales

64. Incorrect statement of the following
(1) Fungi growing on dung are called coprophilous
(2) Fungal cell wall is made up of chitin and polysaccharides
(3) Asexual spores are generally not found in club fungi
(4) Oospores and zygospores are haploid meiospores

65. Cr-Jacob disease in humans is caused by
(1) Free RNA that lacks protein coat
(2) Smallest cellular pathogen, called mycoplasma
(3) Infectious agent with abnormally folded protein
(4) *Contagium vivum fluidum*

66. Possible condition shown by a reproductively eligible female with Down's syndrome is
(1) Gametes of two different types of allosomes
(2) 50% gametes with 24 chromosomes
(3) 50% gametes without autosome
(4) All gametes with two 21st chromosomes

67. The following is not a blood related Mendelian disorder
(1) Phenylketonuria
(2) Thalassemia
(3) Sickle cell anaemia
(4) Haemophilia

68. Australian Marsupials are an example for
(1) Convergent evolution
(2) Homoplasy
(3) Adaptive radiation
(4) Reverse evolution

69. $N_t = N_0 e^{rt}$. Where 'e' is
(1) Intrinsic rate of natural increase
(2) Base of natural logarithm
(3) Population density at time zero
(4) Population density at time T

70. Identify the correct statement from the following
(1) In autoimmunity, the body attacks self cells
(2) Thymus is a lymphoid organ in which both B-cells and T-cells mature
(3) AIDS is a Mendelian disease
(4) Graft rejections are mainly due to MNPs

71. Perigynous flowers are present in
(1) Guava and cucumber
(2) Plum, peach
(3) Rose and Lotus
(4) *Datura* and Chilli

72. Phyllotaxy of *Alstonia*
(1) Opposite
(2) Whorled
(3) Alternate
(4) Leaf mosaic

73. Supporting roots arise from the lower nodes of stem in
(1) Banyan tree
(2) *Monstera*
(3) *Vanda*
(4) Sugarcane

74. Match the following

List-I	List-II
a. Phaeophyceae	I. Chlorophyll a and b
b. Chlorophyceae	II. Chlorophyll a and d
c. Rhodophyceae	III. Chlorophyll a and c
d. Cyanophyceae	IV. Chl – a only

	a	b	c	d
(1)	III	I	II	IV
(2)	III	II	I	IV
(3)	II	I	III	IV
(4)	I	II	III	IV

75. Bulliform cells are present in

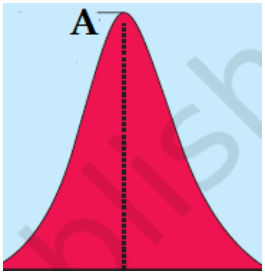
(1) Adaxial side of dorsiventral leaf

(2) Abaxial side of isobilateral leaf

(3) Adaxial side of grass leaf

(4) Abaxial side of monocot leaf

76. Observe the following



Identify the incorrect with respect to the above diagrammatic representation

(1) Stabilizing selection

(2) 'A' – Peak gets higher and narrower

(3) Birth weight is a classic example

(4) Centrifugal selection

77. A person is suffering from internal bleeding, muscular pain, fever, anaemia and blockage of the intestinal passage. He is the sufferer of

(1) Amoebiasis

(2) Filariasis

(3) Ascariasis

(4) Malaria

78. Shortest peptide of proinsulin is

(1) A peptide

(2) B peptide

(3) C peptide

(4) All are equal

79. Which of the following is not true related to sparrow?

(1) Eurythermal animal

(2) Always occupies first trophic level

(3) Has pecten in its eye

(4) Listed as least concern in IUCN Red List

80. Female mosquito

(1) Needs blood meal for survival

(2) Shows incomplete metamorphosis

(3) Needs blood meal for reproduction

(4) Shows external fertilization

81. Assimilatory power consumed in the bundle sheath cells of a C_4 plant for the net gain of one hexose

(1) 30 ATP, 12 NADPH

(2) 12 ATP, 6 NADPH

(3) 18 ATP, 12 NADPH

(4) 12 ATP, 12 NADPH

82. Incorrect statement of the following

(1) Stems are branched in *Pinus* and *Cedrus*

(2) In *Cycas* leaves are pinnately compound and persist for few years

(3) Male and female cones are borne on the same tree in *Pinus*

(4) Male cone and spirally arranged megasporophylls are borne on same tree in *Cycas*

83. An apo enzyme is

(1) Holo enzyme

(2) Zymogen

(3) Protein part of holo enzyme

(4) Co-enzyme

84. During secondary growth of dicot root vascular cambium arises from
(1) Partially from pericycle, partially from medullary rays
(2) Completely from pericycle
(3) Partially from the tissue below protoxylem and remaining from pericycle
(4) Tissue located just below phloem, a portion from pericyle above protoxylem

85. Incorrect match of the following
(1) Protonema – Adult phase of moss plant
(2) Prothallus – gametophyte of pteridophytes
(3) Endosperm – female gametophyte in gymnosperms
(4) Perisperm – persistant nucellus in black pepper

86. Which of the following is not a population interaction?
(1) Commnesalism
(2) Camouflage
(3) Mutualism
(4) Parasitism

87. If 0.6 J of energy is NSP of peacock, the energy found in plant in the form gross primary productivity is? (If rate of respiration in plant is 25% of GPP)
Plant → Mouse → Snake → Peacock
(1) 600 J
(2) 800 J
(3) 200 J
(4) 60 J

88. Regarding threats to biodiversity, the evil quartet is the sobriquet not applicable for
(1) Habitat loss and fragmentation
(2) Over exploitation
(3) Co-evolution
(4) Alien species invasion

89. Kyoto Protocol is concerned with
(1) Reducing the emission of ozone depleting substances
(2) Reducing green house emissions
(3) Reducing the emission of photooxidants
(4) Reducing noise pollution

90. Match the following

Column-I	Column-II
A. *Eicchornia*	I. Hallucinogenic compound
B. *Thylacinus*	II. Anti-cancerous drug
C. *Claviceps*	III. Terror of Bengal
D. *Catharanthus*	IV. Indicators of air pollution
E. Lichens	V. Extinct animal

(1) A-III, B-V, C-I, D-II, E-IV
(2) A-II, B-V, C-I, D-III, E-IV
(3) A-IV, B-V, C-I, D-II, E-III
(4) A-I, B-V, C-III, D-II, E-IV

91. Which of the following is correct ?

(1) Two physical quantities having the same dimension can always be added

(2) Precision always leads to accuracy

(3) A dimensionally correct equation is always correct

(4) If $z = A^a B^b$, then $\frac{\Delta z}{z}$ may not be equal to $\left(a\frac{\Delta A}{A} + b\frac{\Delta B}{B} \right)$

92. A car moving with constant acceleration covers the distance between two points 180 m apart in 6 sec. Its speed as it passes the second point is 45 m/s. What is its acceleration?

(1) -5 m/s^2

(2) -15 m/s^2

(3) 4 m/s^2

(4) 5 m/s^2

93. A projectile is projected from origin of a co-ordinate system with initial velocity of $2\hat{i} + 3\hat{j}\ m/s$ considering y-axis along vertical. The equation of its trajectory is (take g =10 m/s^2)

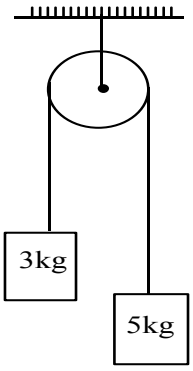
(1) $6y=4x-5x^2$

(2) $5y=6x-4x^2$

(3) $4y= 6x-5x^2$

(4) $6y=5x-4x^2$

94. In the given figure string is massless and pulley is smooth. Find force applied by 3 kg block on the string.



(1) 3g

(2) >3g

(3) <3g

(4) 5g

95. A body of mass 10kg is being acted upon by a force varying with time 't' as (3t^2)N and an opposing constant force of 32N. The initial speed is $10ms^{-1}$. The velocity of body after 5 seconds is

(1) 10.5 m/sec

(2) 6.5 m/sec

(3) 20 m/sec

(4) 4.2 m/sec

96. A lorry and a car moving with same KE are brought to rest by applying the same retarding force. Then

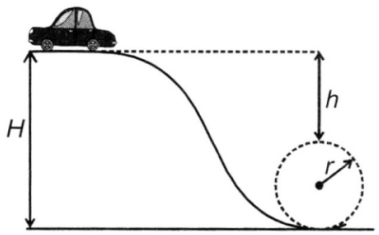
1) Lorry will come to rest in a shorter distance

2) Car will come to rest in a shorter distance

3) Both come to rest in the same distance

4) Data insufficient

97. A toy car slips down the smooth inclined plane as shown in figure. It goes around the vertical smooth circle at the bottom, the relation between H and h is



(1) H = 2h

(2) H = 3h

(3) H = 4h

(4) H = 5h

98. The length of wire is 2.01 m when 5 kg is hanging from it and 2.02 m when 10 kg hanging, then natural length of wire is

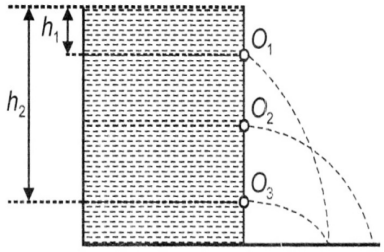
(1) 2m

(2) 1.95m

(3) 1.9m

(4) 2.005m

99. Figure shows a large open tank. Water emerging out from holes O_1 and O_3 strikes the ground at same point but from O_2 has maximum range. The height of O_2 from ground



(1) $\frac{h_1+h_2}{2}$

(2) $\frac{h_2-h_1}{2}$

(3) $\frac{\sqrt{h_1h_2}}{2}$

(4) $2\sqrt{h_1h_2}$

100. Energy required to move a satellite of mass 'm' orbiting around earth of radius R from orbit of radius 2R to orbit of radius 3R is

(1) $\frac{GMm}{12R}$

(2) $\frac{GMm}{3R^2}$

(3) $\frac{GMm}{8R}$

(4) $\frac{GMm}{6R}$

101. A simple pendulum of length *l* and bob of mass m is executing S.H.M with amplitude A. The maximum tension in the string will be

(1) $mg\left(\frac{l}{l+A}\right)$

(2) $mg\left(\frac{l^2+A^2}{l^2}\right)$

(3) $mg\left(\frac{l^2}{A^2+l^2}\right)$

(4) $mg\left(\frac{l+A}{l}\right)$

102. Two sound sources emit the sounds of same frequency f. An observer is at the mid-way between the sources on the line joining them. When he starts moving towards one source with speed v/20, where 'v' is speed of sound, he hears 10 beats per second. The value of 'f' is

(1) 50 Hz

(2) 100 Hz

(3) 150 hz

(4) 200 Hz

103. If a,b,g are the coefficients of linear, areal and volume expansions of a solid then (2a+2b) : (3b+4g) is

(1) 2 : 3

(2) 3 : 2

(3) 1 : 3

(4) 4 : 3

104. Q heat flows per second through the rod of length L and area of cross-section a. If it is stretched to double its length then what will be the heat flow per second keeping temperature difference same?

(1) 2Q

(2) Q/2

(3) Q/4

(4) Q

105. Steam at 100^0C is passed into 55 g of water at 30^0C till the temperature of mixture becomes 90^0C. The mass of the mixture will be

(1) 71g

(2) 61g

(3) 6g

(4) 24g

106. The sink temperature of a Carnot engine of 40% efficiency is at 300 K. For increasing the efficiency to 60%, the increase in the temperature of source temperature will be

(1) 200 K

(2) 300 K

(3) 350 K

(4) 250 K

107. A vessel contains 2 moles of N_2 gas at temperature T. An identical vessel contains 2 moles of Helium gas at temperature 3T. The ratio of their pressure is (Take gases to be ideal)

(1) 1:3

(2) 1:1

(3) 1:27

(4) 27:1

108. RMS speed of a monatomic gas is increased to two times of its initial value. If the process is done adiabatically, then the ratio of final volume to initial volume will be

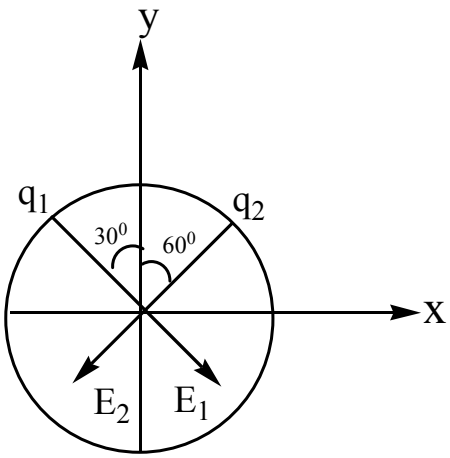
(1) 1:4

(2) 1:8

(3) $1:2^{3/2}$

(4) $1:4^{2/3}$

109. Two charges are arranged as shown in the figure. If electric field at the centre of the circle is along negative y-axis, then $\frac{q_1}{q_2}$ is



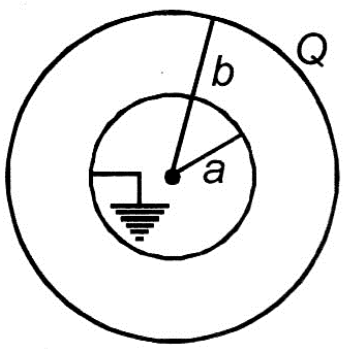
(1) $\sqrt{3}$

(2) $\frac{1}{\sqrt{3}}$

(3) $\frac{1}{3}$

(4) $\frac{\sqrt{3}}{2}$

110. Two concentric spheres of radii a and b as shown in figure are arranged such that outside sphere is charged +Q and inner sphere is earthed. Charge on outer surface of outer sphere is



(1) Q

(2) $Q\left(1+\frac{a}{b}\right)$

(3) $Q\left(1-\frac{a}{b}\right)$

(4) $Q\left(\frac{a}{b}\right)$

111. Three electric dipoles of dipole moment P, 2P, 3P are placed inside a hollow sphere of radius 'a'. The total electric flux coming out of the sphere is

(1) $\frac{6\pi}{a\varepsilon_o}$

(2) *Zero*

(3) $\frac{\pi}{a\varepsilon_o}$

(4) $\frac{3\pi}{a\varepsilon_o}$

112. A parallel-plate capacitor is connected to a battery. The plates are pulled apart with uniform speed. If x is the separation between the plates, then the rate of change of electrostatic energy of the capacitor is proportional to

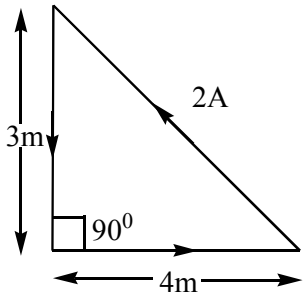
(1) x^2

(2) x

(3) $\frac{1}{x}$

(4) $\frac{1}{x^2}$

113. What is the magnetic moment of the current loop shown below?



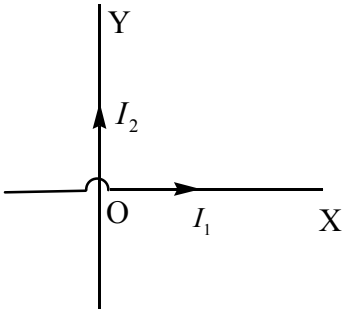
(1) 24 Am^2

(2) 12 Am^2

(3) 6 Am^2

(4) Zero

114. Two long straight conductors with currents I_1 and I_2 are placed along X and Y axes. The locus of points of zero magnetic induction is



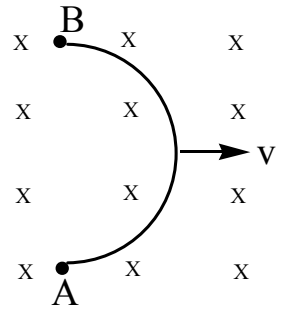
(1) A parabola

(2) A rectangular hyperbola

(3) A straight line

(4) An ellipse

115. A semicircular wire is moving with uniform velocity in a uniform magnetic field as shown in figure. Hence the wire becomes



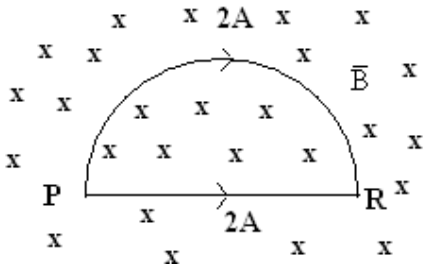
(1) Electrically charged

(2) Positively charged at end B

(3) Hot due to Jouels heating effect

(4) Both (2) & (3)

116. A semi circular current loop is placed in an uniform magnetic field of 1 tesla as shown. If the radius of loop is 1 m, the magnetic force on the loop is



(1) 4N

(2) 8 N

(3) $8/\pi N$

(4) zero

117. The phenomenon of perfect diamagnetism in super conductors is called

(1) Dynamo effect

(2) Meissner effect

(3) Curie effect

(4) Aurora Borealis

118. Two coiles have a mutual inductance of 5 mH. Current changes in the first coil according to the equation $I = I_0 \cos \omega t$ where $I_0 = 10A$ and $\omega = 100\pi \; rad/s$. Maximum value of e.m.f induced in the second coil is

(1) 5π volt

(2) 2π volt

(3) 4π volt

(4) π volt

119. When a coil is connected to a D.C source of emf 12V, a current of 4 amp flows in it. If same coil is connected to 12V, 50Hz AC source, the current is 2.4 A. The self inductance of the coil in henry is

(1) $\frac{1}{20\pi}$

(2) $\frac{1}{10\pi}$

(3) $\frac{1}{25\pi}$

(4) $\frac{1}{5\pi}$

120. Which of the following pairs of space and time varying electric field and magnetic field would generate a plane electromagnetic wave travelling in the z-direction?

(1) $E\hat{i}, B\hat{j}$

(2) $E\hat{i}, B\hat{i}$

(3) $E\hat{i}, B\hat{k}$

(4) $E\hat{k}, B\hat{i}$

121. The intensity of the polarized light becomes $\frac{1}{2a}$ times its initial intensity. The angle between the axis of analyser and polarizer is

(1) $\theta = \tan^{-1}\sqrt{a-1}$

(2) $\theta = \tan^{-1}\frac{1}{\sqrt{a}}$

(3) $\theta = sin^{-1}\sqrt{a-1}$

(4) $\theta = sin^{-1}\frac{1}{\sqrt{a}}$

122. In an ideal Young's double slit experiment using light of wavelength λ, intensity of light at central bright band is I. The intensity of light at point 'P' on the screen where the path difference between the two interfering waves $\frac{\lambda}{6}$ is

(1) I

(2) $\frac{I}{2}$

(3) $\frac{3I}{4}$

(4) $\frac{I}{4}$

123. The difference between (n+1)th Bohr radius and nth Bohr radius is equal to the (n -1) th Bohr radius. The value of n is

(1) 3

(2) 4

(3) 5

(4) 6

124. During an 'α' particle decay from a nuclei at rest of atomic mass 284, the wavelength of the wave associated wih the 'α' particle is 'λ'. Then the wavelength of the wave associated with the daughter nuclei must be

(1) λ

(2) $\frac{\lambda}{2}$

(3) 2λ

(4) 280λ

125. Ninety percent of radioactive sample is left after a time t. The percentage of initial sample that will disintegrate in time 2t is

(1) 38%

(2) 19%

(3) 9%

(4) 62%

126. Two identical particles are projected simultaneously from a tower with speeds V and V/2 at an angle 30^0 and 60^0 simultaneously. The acceleration of centre of mass of the balls when they are in air is

(1) g/3

(2) g/2

(3) g/6

(4) g

127. $\bar{F} = a\hat{i} + 3\hat{j} + 6\hat{k}$ & $\bar{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of 'a' for which angular momentum is conserved is

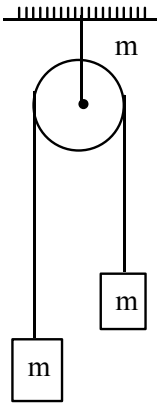
(1) – 1

(2) 0

(3) 1

(4) 2

128. In the system shown, string is light and inextensible but pulley has mass m. Tension in the string connecting pulley with the ceiling is



(1) 2 mg

(2) mg

(3) 3 mg

(4) 4 mg

129. The power developed in an unknown resistance R is same as the power developed in the resistance of 8Ω, when these are conncected separately across a battery of internal resistance 4Ω. The value of R is

(1) 2Ω

(2) 4Ω

(3) 6Ω

(4) 16Ω

130. Potentiometer is most sensitive than voltmeter, because

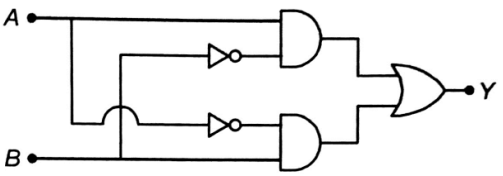
(1) Voltmeter has very high resistance

(2) Least count of potentiometer is small as compared to voltmeter

(3) Potentiometer does not draw any current from a source of unknown potential difference

(4) Potentiometer does not have any coil

131. For the logic circuit given below, the output Y for A = 0, B = 0 and A = 1, B = 1 are



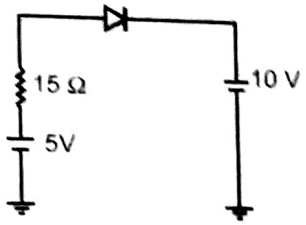
(1) 0 and 1

(2) 0 and 0

(3) 1 and 0

(4) 1 and 1

132. What is the potential difference across the resistor shown in the following circuit diagram if the diode is ideal?



(1) 15 V

(2) 10 V

(3) 5 V

(4) Zero

133. If there is no emergent light through a prism of refracting angle 60^0, whatever may be the angle of incidence, then the minimum value of refractive index of material of the prism is

(1) 2

(2) $\sqrt{2}$

(3) 1.5

(4) $\sqrt{3}$

134. When a ray of light enters from one medium to another then its velocity in second medium becomes doubled. The maximum value of angle of incidence so that total internal reflection may not take place will be

(1) 60^0

(2) 90^0

(3) 30^0

(4) 180^0

135. Focal length of the objective of a terrestial telescope is 80 cm. If it is adjusted for parallel rays, then its magnifyning power is 20.If the focal length of erecting lens is 20 cm, then full length of telescope will be in cm

(1) 84

(2) 100

(3) 124

(4) 164

136. In an emission spectrum of Hydrogen atom transition occurs from n_2 = n to n_1=2 if a graph is plotted between $\bar{\upsilon}(y-axis)$ Vs $\frac{1}{n^2}(x-axis)$, then the y- intercept is equal to

(1) R_H

(2) $\frac{R_H}{4}$

(3) $-R_H$

(4) $\frac{R_H}{2}$

137. Element with highest IP_2 in the periodic table is

(1) Li

(2) Be

(3) B

(4) Na

138. All the following species have bent shape except

(1) SCl_2

(2) ClO_2^-

(3) XeF_2

(4) H_2S

139. 71 grams of 10% pure decane ($C_{10}H_{22}$) on combustion produced _____ grams of CO_2

(1) 220

(2) 22

(3) 2200

(4) 2.2

140. A gaseous mixture containing one gram of hydrogen and "X" grams of Helium taken in 8.21 litre vessel at 227^0C exerted a pressure of 5 atmospheres. "X" is

(1) 1 g

(2) 2 g

(3) 4 g

(4) 0.5 g

141. Consider the formation of the following compounds. In which case $\Delta^0 H_f > \Delta^0 E_f$

(1) CO

(2) CO_2

(3) H_2O

(4) NH_3

142. P^{Ka} of HCOOH is 3.8. P^H of mixture containing two moles of formic acid and two moles of potassium formate is

(1) 4.8

(2) 5.8

(3) 2.8

(4) 3.8

143. $N_2(g) + 3H_2(g) \Leftrightarrow 2NH_3(g), K_C = 100M^{-2}$

Equilibrium constant for the decomposition of 17 grams of ammonia at same temperature is

(1) 10

(2) 0.1

(3) 0.01

(4) 1

144. Which of the following exists as anhydrous salt

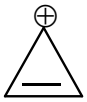
(1) Mg $(NO_3)_2$

(2) Ca $(NO_3)_2$

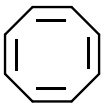
(3) Sr $(NO_3)_2$

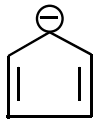
(4) Ba $(NO_3)_2$

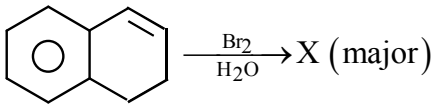
145. Which of the following obey 4 n π electron rule but is non-aromatic due to non planarity

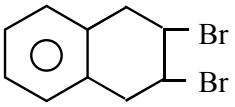
(1) 

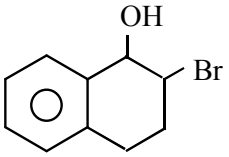
(2) 

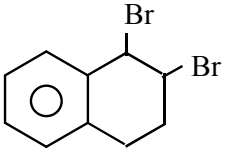
(3) 

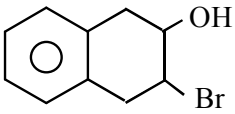
(4) 

146.  . X is

(1) 

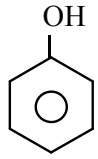
(2) 

(3) 

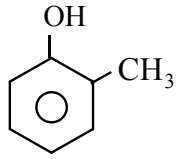
(4) 

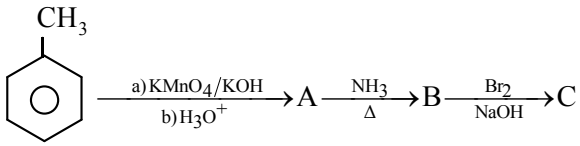
147. P^{Ka} is highest for

(1) $CH_3 - C \equiv CH$

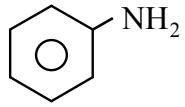
(2) 

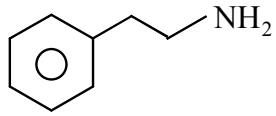
(3) CH_3COOH

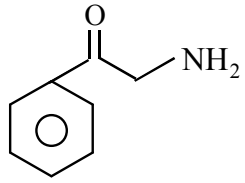
(4) 

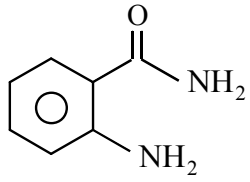
148. 

C is

(1) 

(2) 

(3) 

(4) 

149. Regarding diborane molecule incorrect is

(1) It has six planar atoms

(2) It is a Lewis acid

(3) B - B bond is absent

(4) All B - H bond lengths are equal

150. Incorrect statement is

(1) Silicone rubbers are hydrophobic

(2) Silicone rubbers have low dielectric strength

(3) Quartz is a piezoelectric substance

(4) Quartz is a network solid

151. Which of the following does not liberate oxygen gas on hydrolysis

(1) XeF_4

(2) XeF_2

(3) XeF_6

(4) All of these

152. By product in Solvay ammonia process is

(1) $MgCl_2$

(2) NaCl

(3) $CaCl_2$

(4) $Ca(OH)_2$

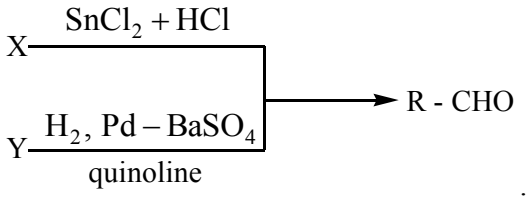
153. Incorrect combination is

(1) Chloroxylenol ……..contains a –OH group

(2) Novestrol ……contains $-C \equiv CH$ group

(3) Aspirin …….. contains -OH group

(4) Codeine …….. contains $-OCH_3$ group

154. 

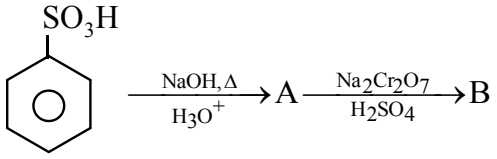
X, Y respectively are

(1) RCN, RCOCl

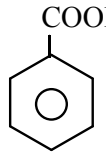
(2) R - CN, RX

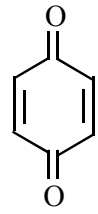
(3) RCH_2NH_2, RCOCl

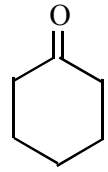
(4) RCN, R_2Cd

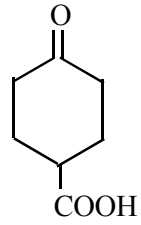
155. 

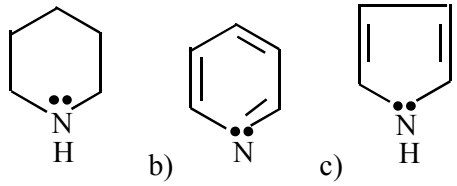
B is

(1) 

(2) 

(3) 

(4) 

156. 

Correct order of basic strength is

(1) a > c > b

(2) b > a > c

(3) c > b > a

(4) a > b > c

157. Wrong match is

(1) $CrO_3 + H_2SO_4$/acetone ……. Jone's reagent

(2) $C_6H_5SO_2Cl$ …….Hinsberg reagent

(3) $KMnO_4 / H^+$ ……. Baeyer's reagent

(4) $[Ag(NH_3)_2]^+$ ……. Tollen's reagent

158. Which of the following is most basic

(1) Alanine

(2) Valine

(3) Glycine

(4) Argenine

159. Given

a) $Zn^{+2} + 2e \rightarrow Zn,\ E^0 = -0.76V$

b) $Na^+ + e \rightarrow Na,\ E^0 = -2.71V$

c) $Ag^+ + e \rightarrow Ag,\ E^0 = +0.80V$

d) $\frac{1}{2}Cl_2 + e \rightarrow Cl^-,\ E^0 = +1.36V$

Correct order of reducing power is

(1) Zn > Na > Ag > Cl^-

(2) Na > Zn > Cl_2 > Ag

(3) Na > Zn > Ag > Cl^-

(4) Cl^- > Ag > Na > Zn

160. Molar mass of $PbSO_4$ is 304g. During discharge of lead storage battery mass of $PbSO_4$ obtained at anode when 0.2F of electricity is generated in

(1) 608 g

(2) 30.4 g

(3) 60.8 g

(4) 304 g

161. Freezing point of 0.1 molal aqueous NaCl which is 100% ionized is (K_f of H_2O = 1.86 K kg $mole^{-1}$)

(1) 0.186^0C

(2) - 0.372^0C

(3) - 3.72^0C

(4) - 0.186^0C

162. Time taken for 99.9% of a first order reaction is 69.3 Min. Rate constant of the reaction is

(1) $10\,min^{-1}$

(2) $0.1\,min^{-1}$

(3) $1\,min^{-1}$

(4) $0.01\,min^{-1}$

163. A) $Al_2(SO_4)_3$ B) $BaCl_2$ C) Na_3PO_4.
The correct decreasing order of coagulating power to cause coagulation in As_2S_3 sol will be

(1) B > A > C

(2) A > C > B

(3) A > B > C

(4) C > B > A

164. Allotropy is not exhibited by

(1) N

(2) P

(3) As

(4) Sb

165. SO_2 is a good reducing agent but it acts like an oxidant in its reaction with

(1) $KMnO_4 / H^+$

(2) $K_2Cr_2O_7 / H^+$

(3) H_2S

(4) Fe^{+3}

166. $Cl_2 + NaOH\ (Hot\ and\ Conc.) \longrightarrow X + Y + H_2O$.
Oxidation states of "Cl" in X and Y respectively

(1) 0, + 3

(2) + 1, + 3

(3) - 1, + 5

(4) + 3, + 5

167. A water sample containing more than 550 ppm of sulphate ion concentration causes

(1) Blue baby syndrome

(2) Kidney failure

(3) Brown mottling of teeth

(4) Laxative effect

168. Hemiacetal formation of glucose involves _____

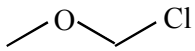
(1) C_1 - C_5 linkage

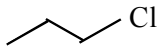
(2) C_2 - C_4 linkage

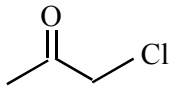
(3) C_1 - C_6 linkage

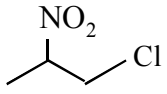
(4) C_2 - C_5 linkage

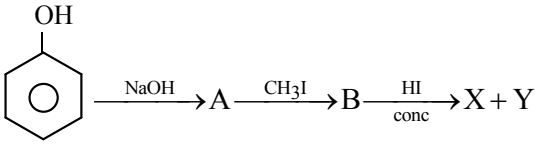
169. Most reactive towards SN^1 reaction is

(1) 

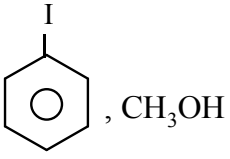
(2) 

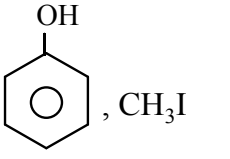
(3) 

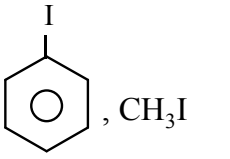
(4) 

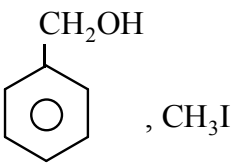
170. 

X, Y are

(1) 

(2) 

(3) 

(4) 

171. Benzene diazonium chloride $\xrightarrow{X}$ C_6H_5I. X is

(1) Cu_2I_2 / HI

(2) Cu / HI

(3) CuI_2

(4) KI

172. Benzene undergoes electrophilic substitution with all the following except

(1) CH_3Cl / $FeCl_3$

(2) Cl_2 / Sunlight

(3) CH_3CH_2COCl / $FeCl_3$

(4) Conc. HNO_3 / Conc. H_2SO_4

173. Which of the following is a sulphate ore

(1) Galena

(2) Zinc blende

(3) Epsom salt

(4) Calemine

174. Enthalpy of atomization is least for

(1) V

(2) Zn

(3) Cr

(4) Fe

175. The element with electronic configuration [Xe] $6s^2 5d^1 4f^7$ is

(1) Gd

(2) Ce

(3) Yb

(4) Th

176. A square planar complex [Mabcd] exhibits _____ isomerism with "X" number of stereo isomers [a, b, c, d are mono dentate ligands]

(1) Optical, 2

(2) Geometric, 2

(3) Optical, 4

(4) Geometric, 3

177. Which of the following is paramagnetic

(1) $Ni(CO)_4$

(2) $[Ni(CN)_4]^{-2}$

(3) $[Zn(NH_3)_4]^{+2}$

(4) $[NiCl_4]^{-2}$

178. Number of radioactive isotopes of hydrogen is/are

(1) 1

(2) 2

(3) 3

(4) zero

179. Volume strength of Perhydrol is

(1) 20 volume

(2) 40 volume

(3) 100 volume

(4) 10 volume

180. An ideal solution is prepared by mixing 1 mole of pure liquid A and 3 moles of pure liquid B. If vapour pressures of pure-A and pure-B are 800 mm and 400 mm respectively. Mole fraction of "B" in vapour phase will be

(1) 0.75

(2) 0.25

(3) 0.5

(4) 0.6

SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

SR ELITE, AIIMS S60, NEET MPL & MEDICON NEET GRAND TEST – 1 KEY Date : 25-01-20

BIOLOGY

1) **3**	2) **2**	3) **1**	4) **3**	5) **3**	6) **3**	7) **4**	8) **3**	9) **2**	10) **3**
11) **1**	12) **4**	13) **2**	14) **3**	15) **4**	16) **3**	17) **4**	18) **1**	19) **4**	20) **4**
21) **4**	22) **4**	23) **3**	24) **1**	25) **4**	26) **3**	27) **3**	28) **3**	29) **2**	30) **2**
31) **2**	32) **3**	33) **1**	34) **4**	35) **2**	36) **4**	37) **1**	38) **2**	39) **1**	40) **1**
41) **2**	42) **3**	43) **2**	44) **4**	45) **4**	46) **2**	47) **2**	48) **1**	49) **3**	50) **2**
51) **3**	52) **4**	53) **4**	54) **4**	55) **4**	56) **3**	57) **1**	58) **3**	59) **4**	60) **2**
61) **2**	62) **2**	63) **1**	64) **4**	65) **3**	66) **2**	67) **1**	68) **3**	69) **2**	70) **1**
71) **2**	72) **2**	73) **4**	74) **1**	75) **3**	76) **4**	77) **3**	78) **1**	79) **2**	80) **3**
81) **3**	82) **4**	83) **3**	84) **4**	85) **1**	86) **2**	87) **2**	88) **3**	89) **2**	90) **1**

PHYSICS

91) **4**	92) **4**	93) **3**	94) **2**	95) **2**	96) **3**	97) **4**	98) **1**	99) **1**	100) **1**
101) **2**	102) **2**	103) **3**	104) **3**	105) **2**	106) **4**	107) **1**	108) **2**	109) **1**	110) **3**
111) **2**	112) **4**	113) **2**	114) **3**	115) **2**	116) **2**	117) **2**	118) **1**	119) **3**	120) **1**
121) **1**	122) **3**	123) **2**	124) **1**	125) **2**	126) **4**	127) **1**	128) **3**	129) **1**	130) **3**
131) **2**	132) **4**	133) **1**	134) **3**	135) **4**					

CHEMISTRY

136) **2**	137) **1**	138) **3**	139) **2**	140) **2**	141) **1**	142) **4**	143) **2**	144) **4**	145) **3**
146) **2**	147) **1**	148) **1**	149) **4**	150) **2**	151) **3**	152) **3**	153) **3**	154) **1**	155) **2**
156) **4**	157) **3**	158) **4**	159) **3**	160) **2**	161) **2**	162) **2**	163) **3**	164) **1**	165) **3**
166) **3**	167) **4**	168) **1**	169) **1**	170) **2**	171) **4**	172) **2**	173) **3**	174) **2**	175) **1**
176) **4**	177) **4**	178) **1**	179) **3**	180) **4**					

SOLUTIONS

PHYSICS

91. Conceptual

92. $v = u + at \Rightarrow 45 = u + 6a$.....(1)

$S = ut + \frac{1}{2}at^2 \Rightarrow 160 = 6u + 18a$....(2) On solving, a = 5ms^{-2}

93. $y = Ax - Bx^2$

$A = \tan\theta = \frac{3}{2}$ and $B = \frac{g}{2(a_x)^2} = \frac{10}{2 \times 2^2} = \frac{5}{4}$

94. $T = m(g + a)$



95. $F = 3t^2 - 32$

$m(v - u) = \int_0^5 (3t^2 - 32)dt \quad \Rightarrow 10(v - 10) = -35 \Rightarrow v = 6.5\ ms^{-1}$

96. Conceptual

97. $\sqrt{2gH} = \sqrt{5gr}$ But, $2r = H - h$

98. $e \propto f \Rightarrow \frac{2.01 - l}{2.02 - l} = \frac{5g}{10g}$

99. To have the same range, $h_1 + h_2 = H$

To have the max range, $h = \frac{H}{2} = \frac{h_1 + h_2}{2}$

100. $E = \Delta TE = \frac{GMm}{2(2R)} - \frac{GMm}{2(3R)} = \frac{Gmm}{12R}$

101. $T_{max} = \frac{mv^2}{l} + mg = \frac{m}{l}(Aw)^2 + mg$

$\Rightarrow \frac{M}{l}\left[A^2 \times \frac{g}{l}\right] + Mg$

$\Rightarrow mg(A^2 / l^2 + 1)$

102. $\Delta n = \left[\frac{v+v_0}{v} - \frac{v-v_0}{v}\right] n = \left[\frac{2v_0}{v}\right] n \Rightarrow n = 100Hz$

103. $\frac{2a+2b}{3b+4g} = \frac{2a+2(2a)}{3(2a)+4(3a)} = \frac{6a}{18a}$

104. $\frac{dQ}{dt} \propto \frac{A}{l} \Rightarrow \frac{dQ}{dt} \propto \frac{V}{l^2} \Rightarrow \frac{dQ}{dt} \propto \frac{1}{l^2}$

105. $(mL + ms\Delta\theta)_{steam} = (ms\Delta\theta)_{water} \Rightarrow m[540 + 1\times 10] = 55\times 1\times 60 \Rightarrow m = 6$

106. $n = 1 - \frac{T_2}{T_1} \Rightarrow \frac{40}{100} = 1 - \frac{300}{T_1} \Rightarrow T_1 = 500K$

$\Rightarrow \frac{60}{100} = 1 - \frac{300}{T_1^1} \Rightarrow T_1^1 = 750K \Rightarrow T_1^1 - T_1 = 250K$

107. $P = \frac{nRT}{V} \Rightarrow \frac{P_1}{P_2} = \frac{2}{2} \times \frac{T}{3T} = \frac{1}{3}$

108. $V_{rms} \propto \sqrt{T} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{T_1}{T_2} = \frac{1}{4}$

$TV^{\gamma-1} = const \Rightarrow \frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$

109. $\tan 30^0 = \frac{E_2}{E_1} = \frac{q_2}{q_1}$

110. $V_{in} = 0 \Rightarrow k\left[\frac{q^1}{a} + \frac{Q}{b}\right] = 0 \Rightarrow q^1 = -\frac{Qa}{b}$ Charge on outer surface = $Q + q^1 = Q - \frac{Qa}{b}$

111. Charge enclosed by sphere is zero

112. $U = \frac{1}{2}CV^2 = \frac{\varepsilon_0 AV^2}{2x} \Rightarrow \frac{dU}{dt} = \frac{\varepsilon_0 AV^2}{2}\left(\frac{-1}{x^2}\right)\frac{dx}{dt} \Rightarrow \frac{dU}{dt} \propto \frac{1}{x^2}$

113. $M = iA = 2\left(\frac{1}{2}\times 4\times 3\right) = 12Am^2$

114. $B_1 = B_2 \Rightarrow \frac{\mu_0 I_1}{2\pi y} = \frac{\mu_0 I_2}{2\pi x} \Rightarrow y = \left(\frac{I_1}{I_2}\right)x$

115. Conceptual

116. $F = f_{curved} + f_{straight} = Bi(2r) + Bi(2r) = 4Bir = 8N$

117. Conceptual

118. $E_s = -M\frac{di}{dt} = +MI_0\omega\sin(\omega t) \Rightarrow E_o = MI_0\omega$

119. $R = \frac{V}{I} = \frac{12}{4} = 3\Omega$ and $Z = \frac{V}{I} = \frac{12}{2.4} = 5\Omega$

$\Rightarrow Z = \sqrt{R^2 + X_L^2} \Rightarrow X_L = 4\Omega \Rightarrow L\omega = 4 \Rightarrow L = \frac{4}{2\pi \times 50} = \frac{1}{25\pi} H$

120. Conceptual

121. $I = \frac{I_0}{2}\cos^2(\theta) \Rightarrow \frac{I_0}{2a} = \frac{I_0}{2}\cos^2\theta \Rightarrow \cos\theta = \sqrt{\frac{1}{a}}$

122. $\phi = \frac{2\pi}{\lambda}(\Delta x) = \frac{\pi}{3}$ *and* $I^1 = I_{\max}\cos^2\left(\frac{\phi}{2}\right) = I\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3I}{4}$

123. $r = (0.53)\frac{n^2}{z} \Rightarrow \left[(n+1)^2 - n^2\right] = (n-1)^2 \Rightarrow n^2 + 1 + 2n - n^2 = n^2 + 1 - 2n$

$\Rightarrow n^2 - 4n = 0 \Rightarrow n = 0$ *or* 4

124. $\lambda = h/p$, Here 'P' is same

125. $\frac{90}{100} = e^{-}\lambda t \Rightarrow \frac{N}{N_0} = e^{-\lambda 2t} = \left(\frac{90}{100}\right)^2 = 0.81 = 81\%$

$\Rightarrow \frac{N^1}{N_0} = 100 - 81 = 19\%$

126. Conceptual

127. Angular momentum is conserved when external torque is zero. Hence $\vec{r} \times \vec{F} = 0$

128. Pulley is in equilibrium

129. $P_1 = P_2 \Rightarrow \left(\frac{E}{R_1 + r}\right)^2 R_1 = \left(\frac{E}{R_2 + r}\right)^2 R_2 \Rightarrow \left(\frac{1}{R+4}\right)^2 R = \left(\frac{1}{8+4}\right)^2 8 \Rightarrow R = 2\Omega$

130. Conceptual

131. $Y = A.\overline{B} + \overline{A}.B$

132. Diode is reverse based hence no current

133. $A = 2\theta_c \Rightarrow \theta_c = 30^0$

$\sin\theta_c = \frac{1}{\mu} \Rightarrow \mu = 2$

134. $\sin\theta_c = \frac{\mu_r}{\mu_d} = \frac{v_d}{v_r} = \frac{v}{2v} \Rightarrow \theta_c = 30^0$

135. $f_0 = 80cm, M = \frac{f_0}{f_e} = 20 \Rightarrow f_e = 4cm$

$L = f_o + 4f + f_e = 80 + 4(20) + 4 = 164cm$